

Network and System Security (TCS 679)



B. Tech CSE-Cyber Security VI Semester

Instructor:

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About the instructor

Qualification:

1. Postdoc from Cyber Security and Networks Lab, **Innopolis University, Innopolis, Russia.**
2. Ph. D (CSE) from Center for Security, Theory and Algorithmic Research of the **International Institute of Information Technology (IIIT), Hyderabad, India.**
3. M. Tech. Computer Network Engineering from **Graphic Era University, Dehradun, India.**
4. B. E. (CSE) from **KEC, Dwarahat, (Regional Engineering College (REC) Uttarakhand, India.**



About the instructor

Research background:

❖ Published more than 200 papers in international journals and conferences in the fields of

Cyber Security

Authentication

Internet of Things (IoT)

Secure Cloud Computing and Edge/fog Computing

ML Security

Big Data

Blockchain

Quantum Cryptography

About the instructor

Research background:

❖ Some of the research findings were published in top cited journals, such as the IEEE TIFS, IEEE TDSC, IEEE Transactions on Smart Grid, IEEE Internet of Things Journal, IEEE Transactions on Industrial Informatics, IEEE Journal of Biomedical and Health Informatics, IEEE Consumer Electronics Magazine, Future Generation Computer Systems, Security and Communication Networks (Wiley) and Journal of Network and Computer Applications.

Awards:

- Rising star of science by Research.com
- University Gold Medal in M. Tech program
- Young Scientist Award by UCOST, Department of Science and Technology, Government of Uttarakhand
- Dr. A.P.J Abdul Kalam innovator of the year award
- IEEE TNSE Excellent Research Reviewer Award for 2022 and 2024
- ICT Express (Elsevier) Best Research Reviewer Award for 2019

Course Outcomes

- After completion of the course students will be able to:
 1. Understand the basics of computer security
 2. Elaborate the cryptographic techniques.
 3. Discuss the transport layer security
 4. Find the pros and cons of various key distribution methods
 5. Analyze the wireless network security
 6. Find the level of system security

Grading scheme

- **As per the University scheme.**
- **Criteria for internal marks:**
 - Performance in the class (queries and answers)
 - Performance in the class tests/quizzes/assignments
 - Attendance, behaviour and activities
- **Criteria for external marks:**
 - Performance in the written exam.

Text/Reference Books:

- Cryptography and Network Security: Principles and Practice by William Stallings (7th edition)
- Ch. P. Pfleeger, S. L. Pfleeger Security in Computing , 4th Edition
Prentice Hall, 2006

Unit-1

- Computer Security Concepts,
- The OSI Security Architecture,
- Security Attacks,
- Security Services,
- Security Mechanisms,
- Models for network security, standards

Unit-1

Cryptography overview

- Cryptography (from Greek: kryptós “secret”; and graphein, “to write”) that is secret writing.
- Cryptography is a mechanism of protecting information through the use of certain codes so that intended recipient can read and process it.
- The generated codes allow information to be kept secret.
- It converts data into a format that is unreadable for an unauthorized user.
- Allowing it to be transmitted without unauthorized entities decoding it back.

Cryptography overview

- It is a method of storing and transmitting data in a particular form so that only those for whom it is intended can read and process it.
- Cryptography not only protects data from theft or alteration, but can also be used for user authentication.

Information security

- Information security is the set of processes that maintain the confidentiality, integrity and availability of the business data in its various forms.
- It is designed to protect the confidentiality, integrity and availability of data from the malicious intentions.
- Confidentiality, integrity and availability are sometimes referred to as the CIA Triad of information security.

Information security

- It is basically the practice of preventing unauthorized access, use, disclosure, disruption, modification, inspection, recording or destruction of information.
- Information can be anything like your details or we can say your profile on social media, your data in mobile phone, your biometrics etc.

Information security

- **Example-**
- During First World War, Multi-tier Classification System was developed keeping in mind sensitivity of information.
- With the beginning of Second World War formal alignment of Classification System was done.
- Alan Turing was the one who successfully decrypted Enigma Machine which was used by Germans to encrypt warfare data.

General security requirements

1. Confidentiality

- This property provides protection of information against any kind of unauthorized access.
- In another way, it is known as “privacy” which protects the exchanged messages against any kind of disclosure attacks.

2. Integrity

- It provides assurance of real and accurate data.
- The content of the received message should not contain illegal insertion, unapproved modification and deletion.
- Data should be safeguarded against unauthorized modification.

General security requirements

3. Authentication

- It is used to validate the identities of the communicating devices and users.
- Mutual verification of identities is mandatory to start a secure communication and it should be conducted in advance.
- Example- In a IoT environment, authentication may occur among the smart IoT devices, various kind of servers (i.e., cloud, fog, edge servers), different users, the service providers.

General security requirements

4. Non-repudiation

- It provides assurance that any entity should not deny the validity of something for example, a message.
- It is one of important service in “information security” which provides proof regarding the “origin of the message” and “integrity of the data” in that message.
- Therefore, it is very difficult for the illegal entities to deny the “**origin of the message**” and “**authenticity of the message**”.

General security requirements

4. Non-repudiation

- Digital signatures methods are useful for “non-repudiation”.
- For example, in case of a online transactions, where it is important to assure that a party to whom the contract was made (communicated party) should not deny the originality of his/her signature on the report.
- **Source's non-repudiation:** It provides assurance of genuineness of the sender. This concludes the message was transmitted by the genuine source.
- **Destination's non-repudiation:** It provides assurance of genuineness of the receiver. This concludes the message was received by the genuine receiver.

General security requirements

6. Forward secrecy

- If an entity (for example, smart home user, smart IoT device) leaves the communication network then it must not have any ability to access the future messages.

7. Backward secrecy

If an entity (for example, smart home user, smart IoT device) is just deployed in the communication network then it must not have any ability to access the previously exchanged messages.

Components of a cryptosystem

1. Plaintext

- It is the data to be protected during transmission.

2. Encryption Algorithm

- It is a mathematical process that produces a ciphertext for any given plaintext and encryption key.
- It is a cryptographic algorithm (i.e., AES) that takes plaintext and an encryption key as input and produces a ciphertext. For example, $E_K(P)=C$, E stands for encryption process, K stands for key used, P stands for plaintext and C stands for produced ciphertext.

Components of a cryptosystem

3. Ciphertext

- It is the scrambled version of the plaintext produced by the encryption algorithm using a specific encryption key.
- The ciphertext is not guarded.
- It is transmitted through a public channel.
- It can be intercepted or compromised by anyone who has access to the communication channel.

Components of a cryptosystem

4. Decryption algorithm

- It is a mathematical process, that produces a unique plaintext for any given ciphertext and decryption key.
- It is a cryptographic algorithm that takes a ciphertext and a decryption key as input, and outputs a plaintext.
- The decryption algorithm essentially reverses the encryption algorithm and is thus closely related to it.
- For example, $D_K(C)=P$, D stands for decryption process, K stands for key used, C stands for ciphertext and P stands for obtained plaintext.

Components of a cryptosystem

5. Encryption key

- It is a value that is known to the sender.
- The sender inputs the encryption key into the encryption algorithm along with the plaintext in order to compute the ciphertext.

6. Decryption key

- It is a value that is known to the receiver.
- The decryption key is related to the encryption key, but is not always identical to it.
- The receiver inputs the decryption key into the decryption algorithm along with the ciphertext in order to compute the plaintext.

Components of a cryptosystem

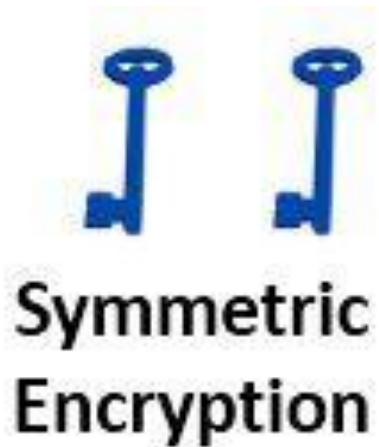
- For a given cryptosystem, a collection of all possible decryption keys is called a key space.
- An interceptor (an attacker) is an unauthorized entity who attempts to determine the plaintext.
- He can see the ciphertext and may know the decryption algorithm.
- He, however, must never know the decryption key.

Symmetric and Asymmetric Encryption

- While communicating on an unsecure channel (i.e., Internet) we have to be careful about the confidentiality of the shared information.
- There are two techniques use to preserve the confidentiality of your message, Symmetric and Asymmetric Encryption.
- Symmetric encryption allows encryption and decryption of the message with the same key.

Symmetric and Asymmetric Encryption

- However, in asymmetric encryption two keys are required. Asymmetric encryption uses the public key for the encryption, and a private key is used for decryption.



Vs



Differences of Symmetric and Asymmetric Encryption

BASIS FOR COMPARISON	SYMMETRIC ENCRYPTION	ASYMMETRIC ENCRYPTION
Basic	Symmetric encryption uses a single key for both encryption and Decryption.	Asymmetric encryption uses a different key for encryption and decryption.
Performance	Symmetric encryption is fast in execution.	Asymmetric Encryption is slow in execution due to the high computational overhead.
Algorithms	DES, 3DES, AES.	Diffie-Hellman, RSA.
Purpose	The symmetric encryption is used for bulk data transmission.	The asymmetric encryption is often used for securely exchanging secret keys.

Definition of Symmetric Encryption

- Symmetric encryption is a technique which allows the use of only one key for performing both the encryption and the decryption of the message.
- It is also known as the conventional method used for encryption.
- In symmetric encryption, the plaintext is encrypted and is converted to the ciphertext using a key and an encryption algorithm.

Definition of Symmetric Encryption

- While the cipher text is converted back to plain text using the same key that was used for encryption, and the decryption algorithm.
- Symmetric encryption algorithm executes faster and is less complex.
- Hence, they are used for bulk data transmission.

Definition of Symmetric Encryption

- In symmetric encryption, the host that are participating in the communication already have the secret key that is received through the external means.
- The sender of the message will use the key for encrypting the message, and the receiver will use the same key for decrypting the message.
- The commonly used symmetric encryption algorithms are DES, 3 DES and AES.

Definition of Symmetric Encryption

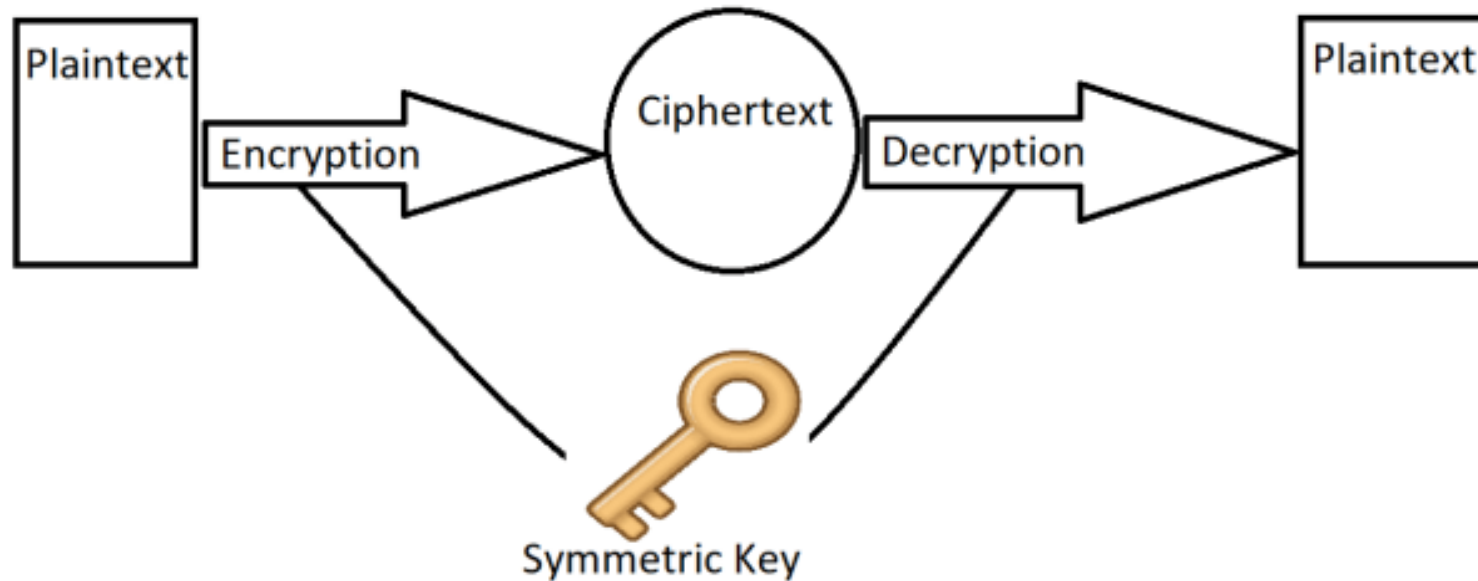


Fig. Procedure of symmetric encryption (source: <https://dzone.com/articles/encryption-and-decryption-using-symmetric-key-in-c>)

Definition of Asymmetric Encryption

- Asymmetric encryption is an encryption technique that uses a pair of key (private key and public key) for encryption and decryption.
- Asymmetric encryption uses the public key for the encryption of the message and the private key for the decryption of the message.
- The public key is available to anyone who is interested in sending the message.

Definition of Asymmetric Encryption

- The private key is kept secret with the receiver of the message.
- Any message that is encrypted by the public key and the algorithm, is decrypted using the same algorithm and the matching private key of corresponding public key.
- The execution of asymmetric encryption algorithm is slow.
- Hence, the asymmetric encryption is used for securely exchanging the keys instead of the bulk data transmission.

Definition of Asymmetric Encryption

- It is generally used for establishing a secure channel over the non-secure medium i.e., Internet.
- The most common asymmetric encryption algorithms are Diffie-Hellman and RSA algorithms.

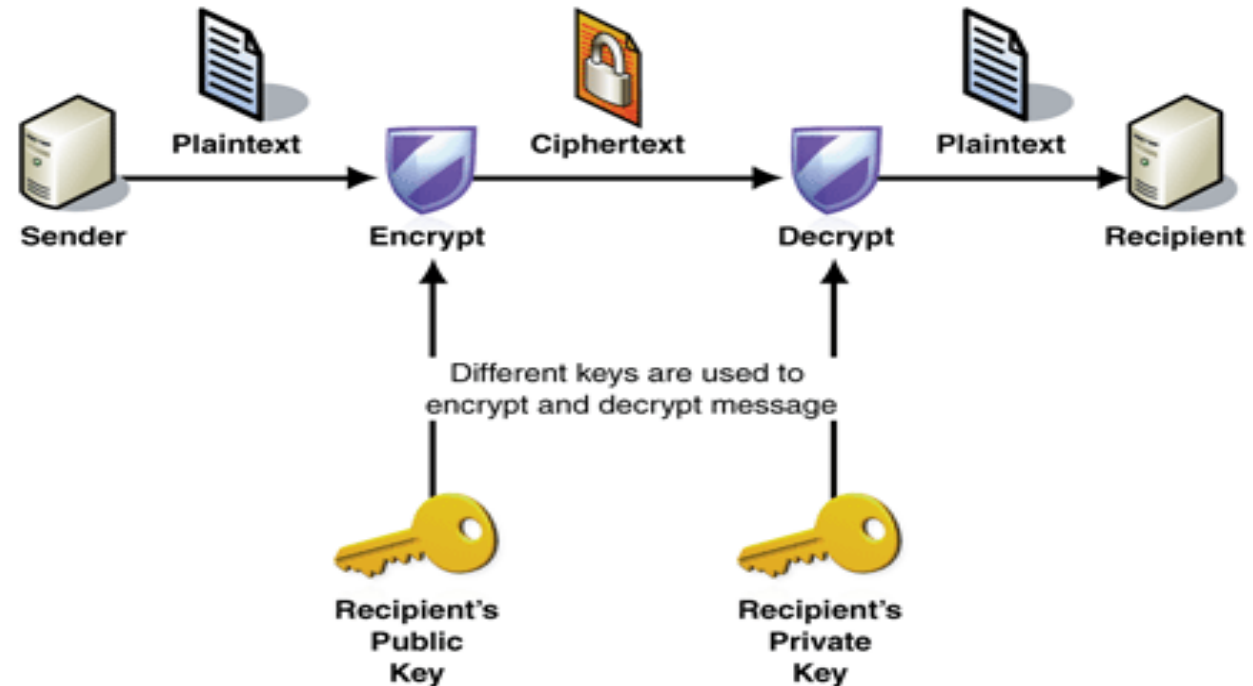


Fig. Procedure of asymmetric encryption
(source:
<https://www.avadaj.com/2017/09/26/combining-symmetric-and-asymmetric-encryption-for-transmitting-data>)

Definition of Asymmetric Encryption

Lesson learned

- Being a complex and slow encryption technique the asymmetric encryption is generally used for exchanging the keys and the symmetric encryption being a faster technique is used for bulk data transmission.

Quiz time

- Choose one correct option.
- Q1: What are the applications of cryptography:
 - (a) Secure Web
 - (b) Secure email
 - (c) Secure electronic transactions
 - (d) All

Quiz time

- Choose one correct option.
- Q2: What are the components of an cryptosystem:
 - (a) Plaintext
 - (b) Ciphertext
 - (c) Encryption and decryption algorithm
 - (d) All

Quiz time

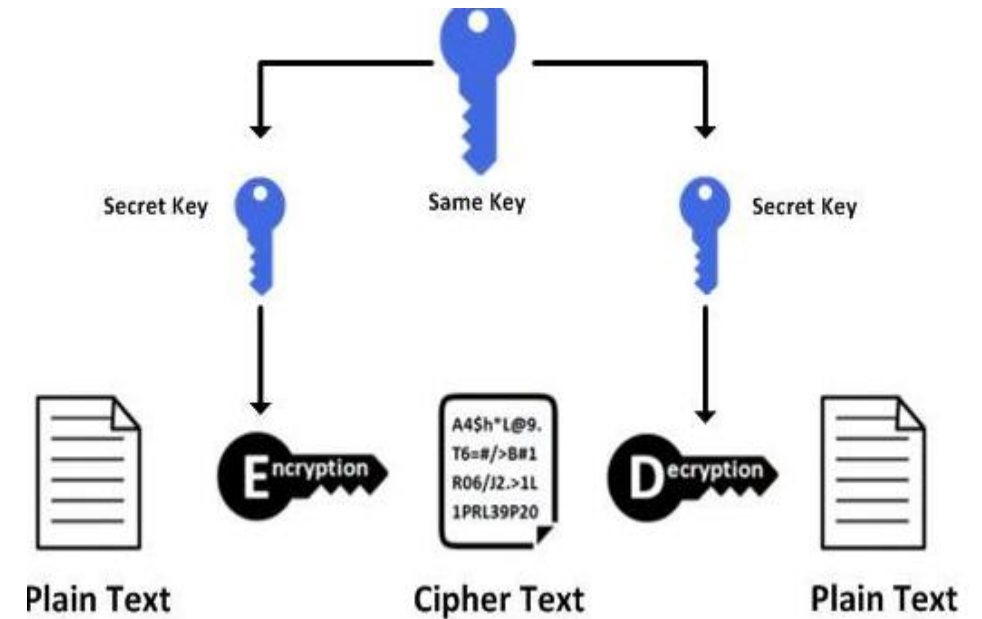
- Choose one correct option.
- Q3: In an asymmetric cryptosystem, we use public key for the encryption and private key for decryption:
 - (a) The above statement is correct
 - (b) The above statement is partially correct
 - (c) Both option a and b are correct
 - (d) None

Quiz time

- Choose one correct option.
- Q4: In an symmetric cryptosystem, we use single key for both encryption and decryption. However, in asymmetric key cryptosystem more than one keys are used.
 - (a) The above statement is correct
 - (b) The above statement is partially correct
 - (c) Both option a and b are false
 - (d) None

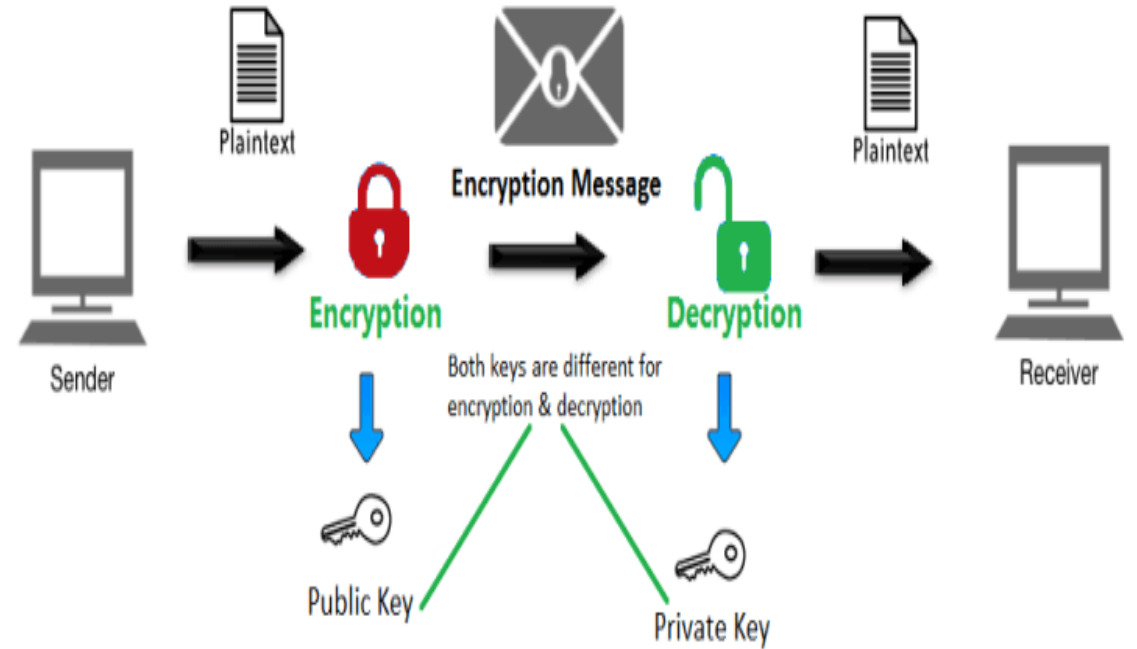
Quiz time

- Choose one correct option.
- Q5: Following figure depicts:
 - (a) A symmetric key cryptosystem
 - (b) A public key cryptosystem
 - (c) None



Quiz time

- Choose one correct option.
- Q6: Following figure depicts:
 - (a) A symmetric key cryptosystem
 - (b) A public key cryptosystem
 - (c) None



Quiz time

- Choose one correct option.
- Q7: A symmetric key cryptosystem executes faster than a public key cryptosystem
 - (a) Correct statement
 - (b) Wrong statement
 - (c) None

References

- ❑ Cryptography and Network Security: Principles and Practice by William Stallings (7th edition)